

ME 310: System Dynamics and Controls

Course Code	ME 310
Course Name	System Dynamics and Controls
L-T-P-C	3-0-0-3
Prerequisites	Engineering Mathematics (IC110) and Linear Algebra (IC111)
Intended for	Discipline Core for ME
Distribution	Core
One-time approval	recommended by the 14th BoA (Sep 2016)

Course Contents

- **Introduction**
Control system examples, historical developments leading to modern day control theory, the basic features and configurations of control systems, control systems analysis and design objectives, control system's design process [2 Lectures]
- **Modeling in Frequency Domain**
Mathematical descriptions of systems by differential equations, Laplace transform linear, time-invariant systems (mechanical, electrical, and electromechanical), linearization of a nonlinear system to find the transfer function. [3 Lectures]
- **Modeling in Time Domain**
State-space representation a LTI system, conversion of a transfer function to state space and vice-versa, linearization a state-space representation. [3 Lectures]
- **Time Response**
Effect of poles and zeros on the time response of a control system, quantitative description of the transient response of first-order systems, general response, damping ratio, settling time, peak time, percent overshoot, and rise time and natural frequency of second-order systems, time response from the state-space representation. [3 Lectures]
- **Reduction of Multiple Subsystems**
Reduction of a block diagram of multiple subsystems to a single block, analysis and design of transient response for a system consisting of multiple subsystems, conversion of block diagrams to signal-flow diagrams, Mason's rule. [2 Lectures]
- **Stability**
Making and interpreting Routh table to determine the stability of a system, Application of Routh table to determine the stability of a system represented in state space. [3 Lectures]
- **Steady-State Errors**
Steady-state error for a unity feedback system, steady-state error performance, design the gain of a closed-loop system to meet a steady-state error specification, finding the steady-state error for disturbance inputs, steady-state error for systems represented in state space. [2 Lectures]
- **Root Locus Techniques**
Properties of a root locus, sketching techniques of a root locus, root locus for systems of order 2 and higher, root locus for positive-feedback systems. [4 Lectures]
- **Design via Root Locus**
Use the root locus to design cascade compensators to improve the steady-state error and the transient response, feedback compensators to improve the transient response, realize the designed compensators physically. [4 Lectures]
- **Frequency Response Techniques**
Plot of frequency response of a system, Nyquist stability criteria, gain and phase margins, closed-loop frequency response. [4 Lectures]
- **Design via Frequency Response**
Use frequency response techniques to adjust the gain to meet a transient response specification, frequency response techniques to design cascade compensators to improve both the steady-state error and the transient response. [5 Lectures]
- **Design via State Space**
State-feedback controller using pole placement, controllability, transient response specifications, observability. [5 Lectures]

Text Books

1. Katsuhiko Ogata, **System Dynamics**, 4th Edition, Pearson Education, 2013
2. N. S. Nise, **Control Systems Engineering**, 7th Edition, Wiley, 2015
3. M. Gopal, **Control Systems: Principle & Design**, 4th Edition, McGraw Hill, 2012

References

1. Franklin, **Feedback Control Systems**, Powell
2. Robert H. Bishop, **The Mechatronics Handbook**.
3. Dean C. Karnopp, Donald L. Margolis and Ronald C. Rosenberg, **Modeling, Simulation, and Control of Mechatronic Systems**, John Wiley & Sons, Inc.
4. Bogdan M. Wilamowski and J. David Irwin, **Control and Mechatronics**, CRC Press
5. William Bolton, **Mechatronics Electronic control systems in mechanical and Electrical Engineering**, 6th Edition, Pearson
6. Richard C. Dorf and Robert H. Bishop, **Modern Control Systems**, Pearson